

Artificial Intelligence Concepts -- Guided Notes

Unit tp-4.6 | CompTIA Network+ N10-009 Obj. FC0-U71 4.6 | N10-008 FC0-U71 4.6

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

AI (Artificial Intelligence) refers to computer systems designed to perform tasks that normally require human _____, such as recognizing speech, identifying images, and making decisions. _____

Learning is a subset of AI where systems learn from _____ without being explicitly programmed for each task.

Question 2

In _____ learning, the training data includes labeled examples -- the model learns from input-output pairs. In _____ learning, the model finds patterns in unlabeled data with no predefined answers.

_____ learning trains an agent to take actions in an environment to maximize a reward signal.

Question 3

A _____ neural network is designed to recognize patterns in image data by scanning the image in overlapping regions. It is used in facial recognition, medical imaging, and self-driving cars. _____

neural networks (RNNs) are designed for _____ data such as text and audio because they retain context from previous inputs.

Question 4

A _____ model (like GPT) predicts the next word in a sequence based on billions of parameters trained on large text datasets. These are also called _____ Models (LLMs). They can generate human-like text, translate languages, write code, and answer questions. A key limitation is that they occasionally produce confident but _____ outputs, called _____.

Question 5

_____ AI refers to systems that excel at one specific narrow task (playing chess, recognizing faces).

_____ AI (AGI) would perform any intellectual task a human can -- it does not yet exist. AI _____ refers to unintended harmful outcomes when AI systems reflect biases present in _____ data.

Question 6

AI _____ is the ability to explain why an AI model made a specific decision. This is important in high-stakes domains like _____, _____, and _____ where decisions must be justifiable to affected individuals. _____ networks are often called black boxes because their decision processes are hard to interpret.

Question 7

Generative AI creates new _____ -- text, images, audio, video, or code -- based on patterns learned from training data. Tools like _____ generate images from text prompts, while _____ generates text. These tools raise ethical concerns about _____ (generating fake but realistic media) and _____.

Question 8 -- Real World Application

REAL WORLD: A school uses an AI tool to screen student applications and rank them. A student advocacy group finds that the model rejects more applications from students at lower-income zip codes, even when their grades are equal to students from higher-income zip codes. Explain what AI problem is occurring and recommend two steps the school should take.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/tp-4-6-artificial-intelligence-concepts/tp-4-6-artificial-intelligence-concepts-answer-key.pdf
```